



Cryptocurrencies & blockchain: the basics for a quant

Blockchain, cryptocurrencies, stablecoins, smart contracts: understanding the building blocks before looking at the risk.

MARKET CAP OF A CRYPTOCURRENCY

Market cap = Unit price × Circulating supply

BLOCKCHAIN

- Distributed ledger, replicated across many nodes, practically impossible to alter after the fact.
- Each block contains a hash of the previous one, guaranteeing the integrity of the history.
- Public (Bitcoin, Ethereum) or private/permissioned depending on the use case.

CRYPTOCURRENCIES

- Bitcoin: positioned as a store of value, with a fixed and scarce supply.
- Ethereum: programmable platform, the foundation of decentralized finance (DeFi).
- High volatility, growing correlation with traditional risk assets.

STABLECOINS

- Cryptocurrencies pegged to a stable asset, most often the US dollar.
- Serve as a bridge between the crypto world and traditional currencies for settlement.
- The key risk is the quality and transparency of the reserves backing them.

SMART CONTRACTS

- Self-executing programs deployed on a blockchain (mostly Ethereum).
- Automate agreements without an intermediary (lending, exchange, on-chain derivative).
- The code is the final word: a security flaw can be exploited irreversibly.

Who uses what?

CRYPTO TRADER

Speculates on or hedges positions in Bitcoin, Ethereum and other digital assets.

BLOCKCHAIN DEVELOPER

Designs and audits smart contracts for DeFi protocols.

RISK MANAGER

Assesses counterparty and custody risk for digital assets.

REGULATOR

Oversees stablecoins and exchange platforms to protect investors.

KEY TAKEAWAYS

1. Bitcoin and Ethereum follow different logics: store of value for one, programmable platform for the other.
2. A stablecoin is only as stable as its reserves are genuinely transparent and sufficient — that's the number one point to watch.
3. A smart contract executes exactly what's coded, flaws included — hence the critical importance of a security audit before deployment.



Comparison table

CRITERION	BITCOIN	ETHEREUM	STABLECOIN
Main role	Store of value	Programmable platform	Stable means of settlement
Underlying tech	Proof of Work blockchain	Proof of Stake blockchain	Backed on an existing blockchain
Volatility	High	High	Very low (if well managed)
Typical use case	Long-term investment	DeFi, smart contracts	Settlement, trading, savings

Real-world examples

BITCOIN AS A STORE OF VALUE

Some institutional investors allocate a small share of their portfolio to Bitcoin as a hedge against currency debasement.

ETHEREUM AND DEFI

A user lends or borrows crypto assets directly through a DeFi protocol, without going through a bank.

USDC FOR SETTLEMENT

An exchange platform uses a stablecoin like USDC to settle transactions quickly, without a traditional bank conversion.

SMART CONTRACT SWAP

A smart contract automatically executes the exchange of two tokens as soon as the defined price conditions are met, with no human intermediary.

Quick glossary

BLOCKCHAIN

Distributed, tamper-proof ledger of transactions, replicated across many nodes.

CONSENSUS (POW/POS)

Mechanism that lets participants agree on the valid state of the blockchain.

WALLET

Tool used to store and manage the keys giving access to digital assets.

SMART CONTRACT

Self-executing program deployed on a blockchain.

DEFI

Decentralized finance: financial services (lending, exchange) without a traditional intermediary.

STABLECOIN

Cryptocurrency whose value is pegged to a stable asset, usually the US dollar.

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